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Models are

stuff

Dimensions

- ❖ Interpretable (lineal & weight) Explainable (non-linear & weights)
- ❖ Complexity (learnable params)
- ❖ Hyperparametrization (tunable params)
- ❖ Scalability (weak: same problem, strong: same compute)
- ❖ Footprint
- ❖ Maintenance & updates

Two families

- ❖ Deterministic (function & prediction)
 - Empirical risk min. & hope for generalization
 - Clust., ANN, RL, ...
- ❖ Probabilistic (distribution & inference)
 - Max. likelihood est., Max posterior est.
 - Mostly statistics

Deterministic

- ❖ Feels more reliable
- ❖ Is less realistic

$$f : \mathbb{R}^D \rightarrow \mathbb{R} \quad / \quad f : \mathbb{R}^D \rightarrow \{C_1, C_2, \dots, C_L\}$$

e.g., $f(x) = w^\top x + w_0$

- ❖ Loss func. or metric

Probabilistic

- ❖ Assume & model noise
- ❖ Multivariate dist.
- ❖ Discriminative $p(y|x)$
- ❖ Generative $p(y,x)$

Lazy

- ❖ Occam's Razor != Your Mind
- ❖ Optimization driven
- ❖ Complexity will come back after you
- ❖ Hyperparameters are not cool

10-fold CV (and others)

- ❖ Cross-validation
 - Leave One Out (cost!)
 - k-fold ($k < n$)
 - 5- $\rightarrow$ 20%, 10- $\rightarrow$ 10%
 - Bias (underfit): Consistently bad
 - Variance (overfit): Inconsistent
- ❖ If possible, leave a test partition out
- ❖ Reliable & Consistent performance estimates
- ❖ Impact assessment
 - Models
 - Hyperparameters
- ❖ Efficient use of data

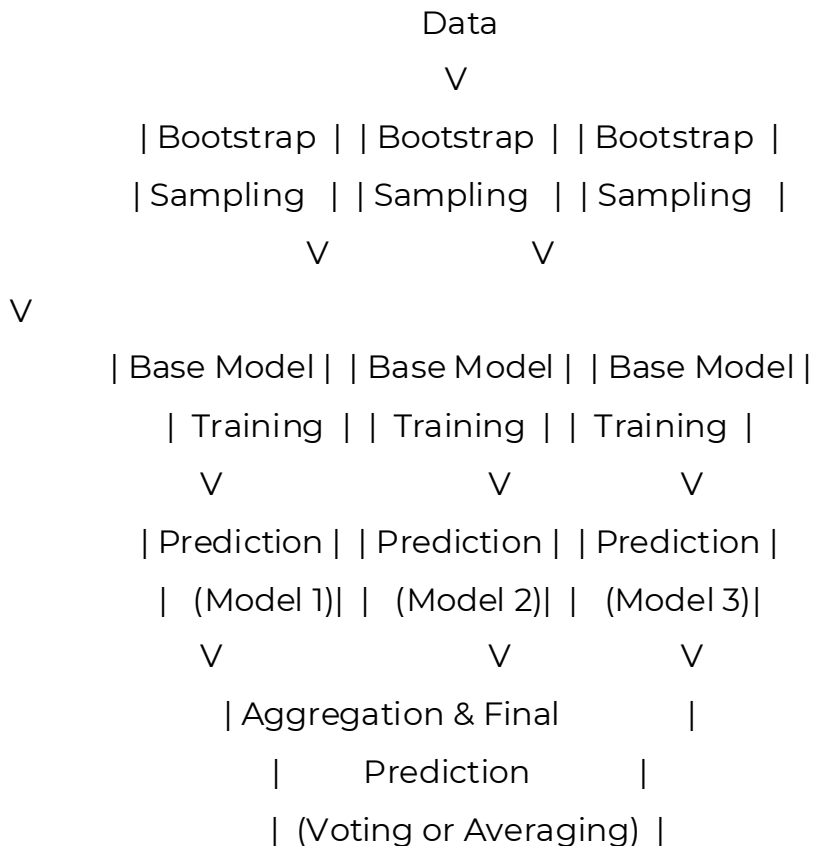
Classic ensembles 1/2

❖ Bootstrapping

- Sample & replace

❖ Bagging

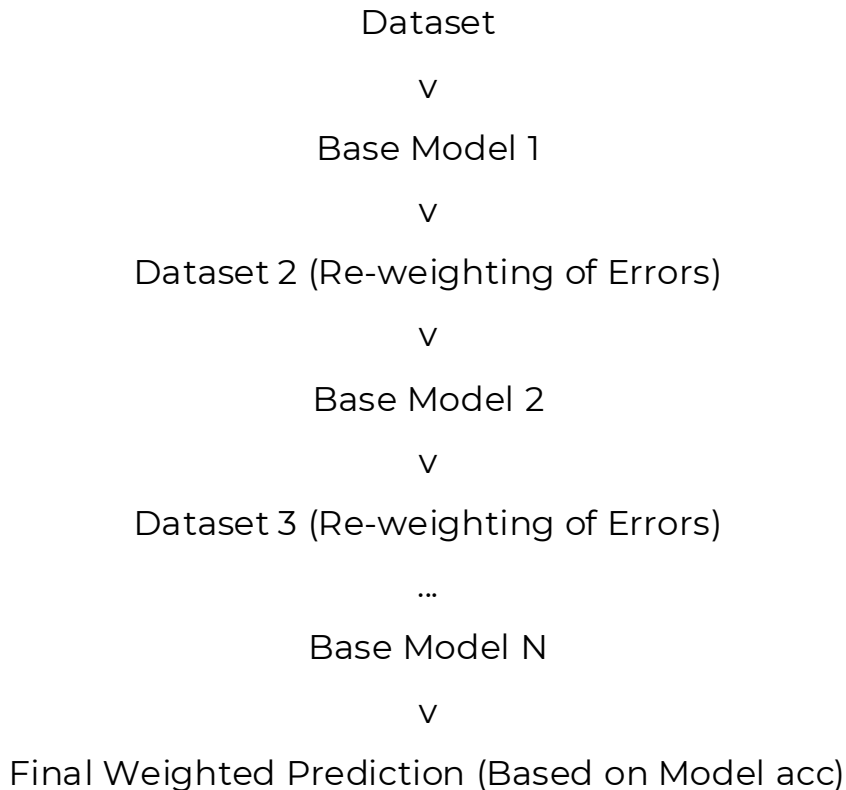
- **Bootstrap & aggregating**
- Final model: Avg./Voting
- Partly overlapping data
- Independent models
- Mixes, and thus reduces, biases and variance



Classic ensembles 2/2

❖ Boosting

- Train a model
- Re-weight mistakes
- Train another model
- Final model: Weighted by acc
- Non-independent models



Experimental approach to model complexity

1. Underfit

- Learn anything (lower bound)

1. Overfit

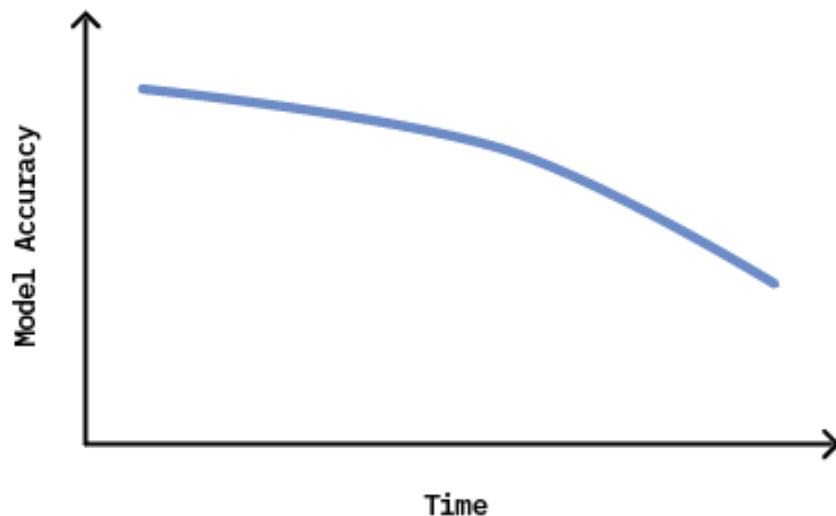
- Learn everything (higher bound)

1. Fit

- Learn the right thing

Models are perishable

- ❖ Concept drift (aka time)
- ❖ Checks
 - Mostly statistics stuff
 - e.g., KS
- ❖ Now also, model collapse
 - Shrinking tails





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Closure

AI's Voldemort

Detoxification

Supervised, how?

Just because we can does not mean we should

Loss of accountability

Normalization of the abnormal

Don't generate. Detect.

Dual use

[]

Molecules

Power & Responsibility

